# Soban Ahmed

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## SUMMARY

Full Stack & AI Engineer with nearly 2 years of production experience building scalable Python backends and RESTful APIs with Django, DRF, and FastAPI, React front ends, voice agents, RAG pipelines, and multi-agent systems on AWS. Hands-on with PostgreSQL, Redis, Celery, Docker, and CI/CD, and has built payments, subscriptions, and enterprise licensing systems; regularly leads client-facing requirement-gathering and translates ambiguous product asks into documented engineering specs.

## EXPERIENCE

**Code Graphers**
*Associate Full Stack Developer* — *Jun 2025 to Present | Gulberg, Lahore*

- Built microservice architecture on AWS ECS with CI/CD pipelines (CodePipeline + CodeBuild + CodeDeploy via CodeCommit) for automated deployments
- Engineered an agentic voice agent using the Pipecat Python SDK, reducing STT latency by 30%
- Built a Twilio-integrated screening bot on Pipecat media streams (replacing Daily.co), cutting infrastructure costs by 50%
- Built a geo-location auto-assignment system and an AI RAG pipeline achieving 90%+ accuracy categorizing store data from real-time captured images

**Speak2Talk, Backend Engineer (Project within Code Graphers, a multi-sided EdTech platform: live classes, AI practice, subscriptions, payouts)**

- Deployed and operated production backend on AWS EC2 with Docker; S3 + CloudFront for media delivery; CI/CD via AWS CodePipeline + CodeBuild + CodeDeploy integrated with CodeCommit
- Built plan-gated subscription/entitlement APIs (curriculum, live class, Sara AI), enterprise licensing, and a Share & Earn affiliate commission flow enabling dual-sided monetization on a shared wallet model
- Designed teacher payroll and payout pipeline (session ledger, period batches, wallet credit, then Stripe Connect withdrawal), automating reconciliation of 200 weekly class-session records previously handled manually
- Implemented request throttling using Redis to protect backend services from abuse and overload
- Self-hosted LiveKit on EC2 for real-time video classrooms, including auth tokens, session lifecycle APIs, and S3-backed recording (Egress) for playback
- Architected Sara AI, a multi-agent RAG practice companion (Chat, Speaking, Pronunciation) using LangGraph, Qdrant vector DB with hybrid search, and Cohere reranking; grounding LLM coaching in a proprietary corpus to eliminate hallucinated grammar/phoneme feedback

**Artilence**
*Software Engineering Intern* — *Aug 2024 to Dec 2024 | Johar Town, Lahore*

- Built scalable RESTful APIs with Django DRF and FastAPI, backed by PostgreSQL and Redis, for an AI-driven multi-tenant ATS powered by OpenAI and VAPI
- Engineered a real-time student community platform using Golang (Fiber) and WebSockets with media chunking, compression, and Alembic migrations; prototyped RAG/LangGraph multi-agent pipelines

## PROJECTS

**AI-Powered Project Management Tool** — *Python, Django, React, WebSockets, WebRTC, LangChain, PostgreSQL*

Built a multi-tenant project management platform (JWT auth, RBAC, Kanban, analytics dashboard); developing LangChain-powered task summarization/auto-creation

**AI Mock Interview Platform** — *FastAPI, React, PostgreSQL, VAPI*

Built a secure FastAPI backend with JWT auth and LLM-powered resume parsing; used VAPI for voice-based AI interviews with structured feedback

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, SQL, Golang, C++, Bash

**Frameworks:** Django, Django REST Framework, FastAPI, React, Vue.js, Tailwind CSS, LangChain, LangGraph

**AI/ML:** RAG Pipelines, Multi-Agent Systems, LLM Integration, Vector Databases (Qdrant), Agentic RAG, OpenAI API, VAPI, Pipecat, Deepgram

**Infrastructure:** AWS (EC2, ECS, S3, CloudFront, CodeCommit, CodeBuild, CodeDeploy), Docker, CI/CD, Microservices, RESTful APIs, JWT, PostgreSQL, Redis, Celery, Stripe/Stripe Connect, LiveKit, WebSockets/WebRTC

## EDUCATION

**University of Management and Technology** — Lahore, Punjab
*B.S. in Computer Science, CGPA: 3.2+* — *Fall 2021 to 2025*

- Led competition management for UMT ACM chapter; organized LeetCode coaching sessions and hosted programming events